Advanced FFT Algorithms and Applications

Signals and Linear Systems

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Outline

- **Advanced FFT Algorithms**
  - Radix-2 & Radix-3 DIT Tabular Interpretation
  - General Cooley-Tukey Matrix Formulation
- **Applications of FFT:** Fast Convolution
  - Circular Convolution
  - Linear Convolution via Zero-Padding

Advanced Matrix Interpretations of FFT

Revisiting Radix-2: A Tabular View

- Previously, we derived Radix-2 DIT purely algebraically.
- We divided the input into even and odd indices and calculated their DFTs $G[k]$ and $H[k]$.
- Then we combined them using the butterfly operation:

$$X[k] = G[k] + W_N^k H[k]$$
$$X[k + N/2] = G[k] - W_N^k H[k]$$

- We can think about this in a different, more visual way—one that leads to an elegant generalization.
- **Key Idea:** Reshape the 1D input $x[n]$ into a 2D matrix, perform DFTs along rows and columns.

Radix-2 DIT: Column-Major Input & Row DFTs

- Let us arrange $x[n]$ into a $2 \times N/2$ matrix in **column-major** order:

$$n = 2n_2 + n_1$$

Where,

$$n_1 = \text{row index} \in \{0, 1\}$$
$$n_2 = \text{column index} \in \{0, 1, \ldots, N/2 - 1\}$$

- **Row 0** ($n_1 = 0$): Even indices
- **Row 1** ($n_1 = 1$): Odd indices
- **1st DFT:** Then perform N_2-point DFTs along each row ($\rightarrow$).
  - Row 0 $\rightarrow G[k]$
  - Row 1 $\rightarrow H[k]$

Column-Major Input $x[n]$

	$n_2 = 0$	$n_2 = 1$	$n_2 = 2$	$\cdots$	
$n_1 = 0$ (Even)	$x[0]$	$x[2]$	$x[4]$	$\cdots$	$\rightarrow$ DFT
$n_1 = 1$ (Odd)	$x[1]$	$x[3]$	$x[5]$	$\cdots$	$\rightarrow$ DFT

After Row-wise DFT

	$n_2 = 0$	$n_2 = 1$	$n_2 = 2$	$\cdots$
$n_1 = 0$ (Even)	$G[0]$	$G[1]$	$G[2]$	$\cdots$
$n_1 = 1$ (Odd)	$H[0]$	$H[1]$	$H[2]$	$\cdots$

1st DFTs $N/2$-pt (row-wise)

Radix-2 DIT: Twiddle Factors & Column DFTs

- After row DFTs, multiply row n_1 by twiddle factors $W_N^{n_1 k}$.
  - Row 0: $\times W_N^0 = 1$
  - Row 1: $\times W_N^k$
- **2nd DFT:** Perform 2-point DFT down each column $(\downarrow)$.
- This is exactly the butterfly!

$$X[k] = G[k] + W_N^k H[k]$$
$$X[k+\tfrac{N}{2}] = G[k] - W_N^k H[k]$$

- We obtain the DFT of the original sequence $x[n]$!
- Row 0 gives $X[0\ldots N/2-1]$,
  row 1 gives $X[N/2\ldots N-1]$.
- But the output occurs in **row-major** order.

After Twiddle Factors

	$n_2=0$	$n_2=1$	$n_2=2$	$\cdots$
$n_1=0$ (Even)	$G[0]$	$G[1]$	$G[2]$	$\cdots$
$n_1=1$ (Odd)	$H[0]W_N^0$	$H[1]W_N^1$	$H[2]W_N^2$	$\cdots$

DFT DFT DFT DFT

After Column-wise DFT

	$n_2=0$	$n_2=1$	$n_2=2$	$\cdots$
$n_1=0$ (Even)	$X[0]$	$X[1]$	$X[2]$	$\cdots$
$n_1=1$ (Odd)	$X[N/2]$	$X[1+N/2]$	$X[2+N/2]$	$\cdots$

Radix-2 DIT: Twiddle Factors & Column DFTs

- We want $X[k]$ in the correct order. We want the transpose matrix.
- Mathematically, for

$$\begin{aligned} k_1 &= \text{row index} \in \{0, 1\} \\ k_2 &= \text{column index} \in \{0, \ldots, N/2 - 1\} \end{aligned}$$

- We want the **row-major** indexing:

$$k = (N/2)k_1 + k_2$$

- Using this index, we can write:

$$X[(N/2)k_1 + k_2] = G[k_2] + W_2^{k_1} \left(W_N^{k_2} H[k_2] \right)$$

- Not the final form yet!
- But **key take away**: We can visualize Cooley-Tukey algorithm as arranging numbers on a matrix, performing DFTs on columns and rows!

After Output Indexing

	$k_2 = 0$	$k_2 = 1$	$k_2 = 2$	$\cdots$
$k_1 = 0$ (Even)	$X[0]$	$X[1]$	$X[2]$	$\cdots$
$k_1 = 1$ (Odd)	$X[0 + N/2]$	$X[1 + N/2]$	$X[2 + N/2]$	$\cdots$

Radix-3 DIT DFT: Derivation

- The same idea extends naturally. For $N = 3N_2$, split $x[n]$ into three subsequences by index modulo 3:

$$X[k] = \sum_{r=0}^{N/3-1} x[3r]\, W_{N/3}^{rk} + W_N^{k} \sum_{r=0}^{N/3-1} x[3r+1]\, W_{N/3}^{rk} + W_N^{2k} \sum_{r=0}^{N/3-1} x[3r+2]\, W_{N/3}^{rk}$$

- Define three $N/3$-point DFTs:

$$G_m[k] = \sum_{r=0}^{N/3-1} x[3r+m]\, W_{N/3}^{rk} \qquad m \in \{0, 1, 2\}$$

Radix-3 DIT Combination

$$\begin{aligned} X[k] &= G_0[k] + (W_N^{k}\, G_1[k]) + (W_N^{2k}\, G_2[k]) \\ X[k+N/3] &= G_0[k] + W_3^{1}(W_N^{k}\, G_1[k]) + W_3^{2}(W_N^{2k}\, G_2[k]) \\ X[k+2N/3] &= G_0[k] + W_3^{2}(W_N^{k}\, G_1[k]) + W_3^{4}(W_N^{2k}\, G_2[k]) \end{aligned}$$

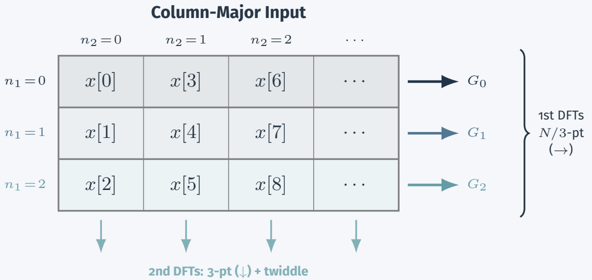
- The above 3 equations are also a 3-point DFT!

Radix-3 DIT: Tabular Interpretation ($N_1 = 3, N_2 = N/3$)

- Load $x[n]$ into a $3 \times N/3$ matrix, **column-major**: $n = 3n_2 + n_1$.
- **Row 0**: $n \equiv 0 \pmod 3$
- **Row 1**: $n \equiv 1 \pmod 3$
- **Row 2**: $n \equiv 2 \pmod 3$

1. **1st DFTs:** $N/3$-point DFT along each row ($\rightarrow$).
2. Multiply by twiddle $W_N^{n_1 k_2}$.
3. **2nd DFTs:** 3-point DFT down each column ($\downarrow$).

- Take output **row-major**: $k = (N/3)\,k_1 + k_2$.



General Cooley-Tukey Factorization ($N = N_1 N_2$)

- Having seen the tabular view for Radix-2 and Radix-3, let's formalize the general $N_1 \times N_2$ factorization.
- Re-index the input n as an $N_1 \times N_2$ matrix in **column-major order**:

$$n = N_1 n_2 + n_1 \qquad \begin{cases} n_1 \in \{0, 1, \ldots, N_1 - 1\} \\ n_2 \in \{0, 1, \ldots, N_2 - 1\} \end{cases}$$

  Substituting this into the DFT definition yields a 2D sum:

$$X[k] = \sum_{n_1=0}^{N_1-1} \sum_{n_2=0}^{N_2-1} x[N_1 n_2 + n_1] e^{-\frac{2\pi i}{N}(N_1 n_2 + n_1)k}$$

- Re-index the output k as an $N_1 \times N_2$ matrix in **row-major order**:

$$k = N_2 k_1 + k_2 \qquad \begin{cases} k_1 \in \{0, 1, \ldots, N_1 - 1\} \\ k_2 \in \{0, 1, \ldots, N_2 - 1\} \end{cases}$$

General Cooley-Tukey Factorization ($N = N_1 N_2$)

- Substitute $k = N_2 k_1 + k_2$ into the exponent $(N_1 n_2 + n_1)k$:

$$\begin{aligned}(N_1 n_2 + n_1)(N_2 k_1 + k_2) &= N_1 N_2 n_2 k_1 + N_1 n_2 k_2 + N_2 n_1 k_1 + n_1 k_2 \\ &= N n_2 k_1 + N_1 n_2 k_2 + N_2 n_1 k_1 + n_1 k_2\end{aligned}$$

- The first term's exponential is $e^{-\frac{2\pi i}{N} N n_2 k_1} = 1$, so this cross term vanishes.
- We separate the remaining terms in the exponential $e^{-\frac{2\pi i}{N}(\dots)}$:

$$e^{-\frac{2\pi i}{N_2} n_2 k_2} \cdot e^{-\frac{2\pi i}{N_1} n_1 k_1} \cdot e^{-\frac{2\pi i}{N} n_1 k_2}$$

- Substituting this back into the double sum and rearranging:

Cooley-Tukey 2D Matrix Formulation

$$X[N_2 k_1 + k_2] = \sum_{n_1=0}^{N_1-1} \underbrace{\left[e^{-\frac{2\pi i}{N_1 N_2} n_1 k_2} \right]}_{\text{Twiddle Factor}} \left(\sum_{n_2=0}^{N_2-1} x[N_1 n_2 + n_1] e^{-\frac{2\pi i}{N_2} n_2 k_2} \right) e^{-\frac{2\pi i}{N_1} n_1 k_1}$$

- **Inner Sum:** An N_2-point DFT on the rows of the $N_1 \times N_2$ input matrix.
- **Twiddle Factors:** The bracketed term scales the intermediate results by $e^{-\frac{2\pi i}{N_1 N_2} n_1 k_2}$.
- **Outer Sum:** An N_1-point DFT down the columns of the intermediate matrix.

- The Cooley-Tukey matrix formulation directly yields **Bailey's FFT** (a.k.a. the Four-Step FFT):

Bailey's FFT

1. Arrange input $x[n]$ in an $N_1 \times N_2$ matrix (column-major).
2. Compute N_2-point DFT of each row ($\rightarrow$).
3. Multiply entry (n_1, k_2) by twiddle factor $W_N^{n_1 k_2}$.
4. Compute N_1-point DFT of each column ($\downarrow$).

- How to compute the N_1- and N_2-point DFTs? **Apply the same algorithm recursively!**
- If $N = p_1^{a_1} p_2^{a_2} \cdots$, we recursively factor until only small primes remain $\Rightarrow O(N \log N)$ for any composite N.
- This handles powers of 2, powers of 3, mixed radices—any N with small prime factors.
- But what if N itself is a **large prime**?